

Paper VII — Surge-Tolerant Locked Tray with Integral Relief (AEGIS / STT): A Cross-Flow Tray Rated to 5 psi (34.5 kPa) Vapour Uplift Without Loss of Normal Performance

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Invention Disclosure / Provisional Specification Draft — HyTray Series HT-07

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Abstract

A cross-flow contacting tray is disclosed that retains the hydraulic and mass-transfer performance of a conventional valve or sieve tray during normal operation while surviving a transient vapour uplift of up to 5 psi (34.5 kPa) — roughly 20–30 times the uplift resistance of a conventionally clamped tray — without panel dislodgement, buckling or collapse. Three cooperating features achieve this: (i) positive interlocking hook clamps that lock each deck panel to a welded support ring, carrying the full uplift in tension rather than relying on gravity and friction; (ii) close-pitch integral stiffening ribs that keep deck bending stress below allowable under the 5 psi distributed load; and (iii) optional integral relief flaps that crack at a set pressure (e.g. 1.5–2 psi) to vent the surge and cap the *sustained* deck load, so that the locked structure need only survive the brief peak. During normal operation the relief flaps are seated and sealed by their own weight plus liquid head, so the operating window, pressure drop, efficiency and turndown are identical to the underlying valve/sieve deck. Mechanical calculations are given parametrically and for two embodiments (1.2 m and 3.0 m diameter), including deck bending stress versus rib spacing, total uplift force, hook-clamp and support-ring loads, and relief-flap sizing.

Status note: stress and load values are from classical plate/beam theory with stated assumptions (simply supported strips, static-equivalent loads with a dynamic amplification factor). They are design calculations for disclosure, not a substitute for finite-element analysis and code-stamped mechanical design (ASME VIII / EN 13445). Tray hold-down clamps and surge-resistant internals exist; the asserted inventive matter is the integrated lock-plus-relief system that bounds the structural duty.

1 Field of the Invention

Mechanical design of cross-flow distillation/absorption trays for service subject to severe vapour uplift transients (pressure surges, relief events, slug flow, mis-sequenced steam-out, rapid depressurisation).

2 Background and Prior Art

Conventional trays are held against uplift by gravity, by light friction clamps, or by integral self-locking valve units, and are typically rated for an uplift of only about 1–2 kPa (0.15–0.3 psi). This suffices for normal operation, where net uplift is of the order of the dry pressure drop. It does not suffice for upset events: a sudden vaporisation, a relief-valve lift, a liquid slug striking a tray, or an out-of-sequence steam-out can impose uplift an order of magnitude higher, dislodging or buckling

trays. Tray collapse is a recognised cause of column damage and unplanned shutdowns and, in cascade, of mechanical destruction of the entire tray stack.

Existing countermeasures are partial. Heavy hold-down bars and bolting add uplift capacity but make the deck non-removable through the manway and add weight; they also do not address deck-plate buckling under a large distributed pressure. No commercial tray known to the inventor combines (a) positive panel interlock sized for several psi of uplift, (b) deck stiffening sized for the same distributed load, and (c) an integral pressure-relief mechanism that bounds the sustained load while preserving normal tray hydraulics.

3 Objects of the Invention

1. To provide a removable, manway-installable tray that survives 5 psi vapour uplift without dislodgement or buckling.
2. To carry uplift in a defined load path: deck $\rightarrow$ ribs $\rightarrow$ hook clamps $\rightarrow$ welded support ring $\rightarrow$ shell.
3. To bound the sustained deck load by integral relief, so the structure is feasible and light.
4. To leave normal-operation hydraulics, efficiency and turndown unchanged.

4 Summary of the Invention

Each deck panel (84) is locked to a welded support ring (80) by interlocking hook clamps (82) that engage under the ring and resist uplift in tension. The panel underside carries close-pitch stiffening ribs (86) (formed channels or welded flat bars) that reduce the unsupported bending span. One or more panels carry integral relief flaps (88): hinged or weight/spring-loaded covers over enlarged openings that remain seated and sealed during normal operation and crack open when the net upward differential pressure exceeds a set value (e.g. 1.5–2 psi), venting vapour and capping the sustained deck load; the flaps reseal when the surge passes. The locked, stiffened structure is sized to survive the brief 5 psi peak before the flaps fully relieve.

5 Brief Description of the Drawings

Figure 1: tray cross-section showing welded support ring (80), interlock hook clamp (82), deck panel (84), stiffening ribs (86), and relief flap (88) (closed at left, open under surge at right). Figure 2: (a) deck bending stress versus rib spacing at 5 psi for three deck thicknesses; (b) total uplift force versus column diameter; (c) relief-flap load-capping characteristic.

6 Detailed Description and Calculations

1. Total uplift force (parametric). For uplift pressure q over tray area $A = \pi D^2/4$, the total uplift is $F = q \pi D^2/4$. At $q = 5 \text{ psi} = 34,474 \text{ Pa}$:

$$D = 1.2 \text{ m} : F = 34,474 \times 1.131 = 39.0 \text{ kN (3.97 t)}; \quad D = 3.0 \text{ m} : F = 243.7 \text{ kN (24.8 t)}.$$

This force, two-to-three orders above normal hold-down duty, defines the load path (Fig. 2b).

2. Deck bending stress and rib pitch (parametric). Treat the deck between ribs as a simply-supported strip of unit width and span s (the rib pitch) under uniform pressure q . Maximum moment $M = qs^2/8$ per unit width; bending stress in a plate of thickness t :

$$\sigma = \frac{6M}{t^2} = \frac{0.75 q s^2}{t^2}$$

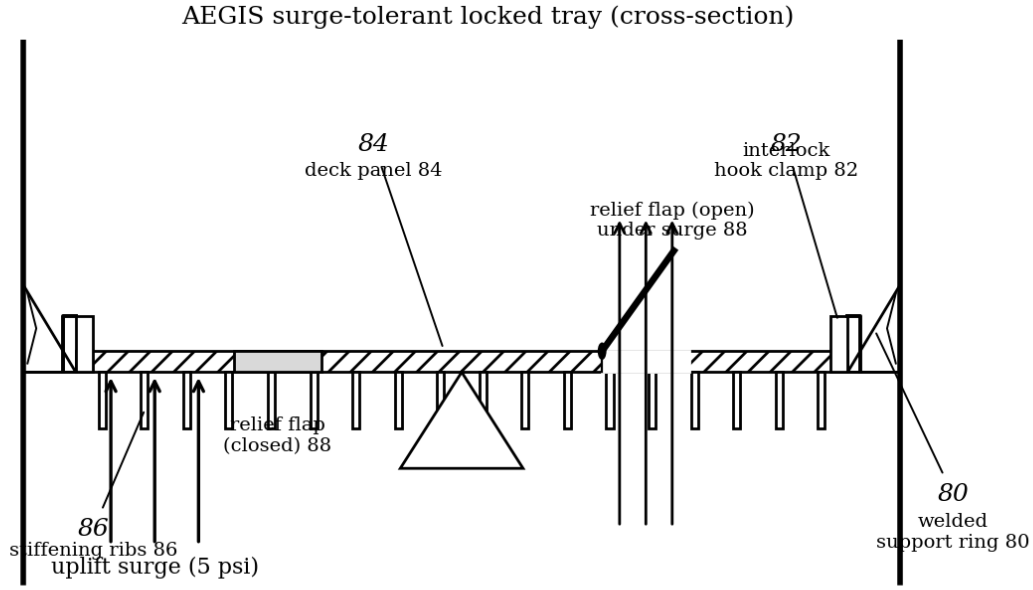


Figure 1: AEGIS tray: support ring (80), interlock hook clamp (82), deck panel (84), stiffening ribs (86), relief flap (88).

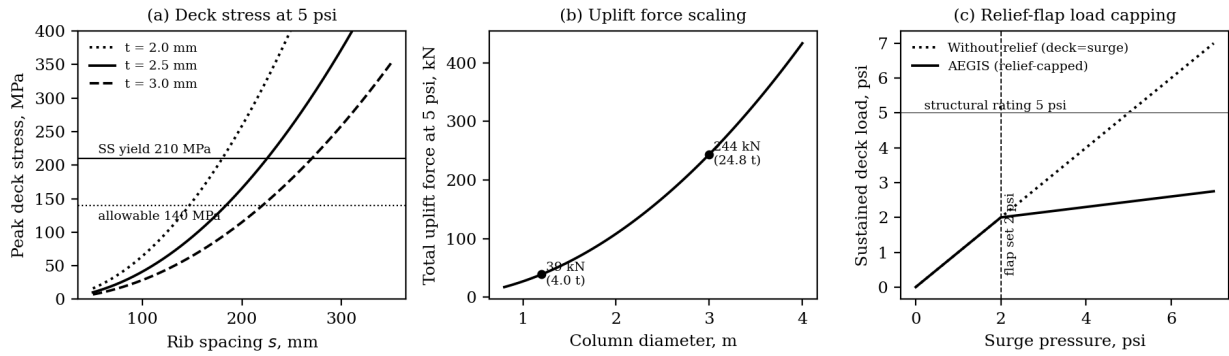


Figure 2: Mechanical design charts (classical theory): (a) deck stress vs rib pitch; (b) uplift force scaling; (c) relief flap capping sustained load while the structure survives the 5 psi peak.

Requiring $\sigma \leq \sigma_{allow}$ gives the maximum rib pitch

$$s_{max} = t \sqrt{\frac{\sigma_{allow}}{0.75 q}}.$$

Embodiment: SS deck, $\sigma_{allow} = 140$ MPa (safety factor ~ 1.5 on 210 MPa yield), $q = 34,474$ Pa:

$$t = 2.5 \text{ mm} : s_{max} = 0.0025 \sqrt{140 \times 10^6 / (0.75 \times 34474)} = 184 \text{ mm}; \quad t = 3.0 \text{ mm} : s_{max} = 221 \text{ mm}.$$

Thus ribs at 150–180 mm pitch on a 2.5 mm deck hold a flat 5 psi load within allowable (Fig. 2a); a bare 2.5 mm panel on a 400 mm span would see $\sigma \approx 662$ MPa and fail, which is why conventional decks buckle under surge.

3. Deck deflection. Strip deflection $\delta = 5qs^4/(384EI)$ with $I = t^3/12$ per unit width. At $s = 170$ mm, $t = 2.5$ mm, $E = 193$ GPa: $\delta \approx 0.9$ mm ($\approx s/190$), within $s/150$ serviceability.

4. Hook-clamp and support-ring load (parametric). The uplift on a panel of width b and chord length L_p is $F_p = qbL_p$, reacted at the two ring edges; clamp tensile load per unit length of engaged ring is $w_c = F_p/(2L_p) = qb/2$. For $b = 0.4$ m: $w_c = 34,474 \times 0.4/2 = 6,895$ N/m. Hook

clamps at 200 mm pitch each carry ≈ 1.38 kN — a modest bolt/hook duty. The welded support ring carries the total F in shear/bending to the shell; ring weld throat sized to $F/(\pi D)$ per unit length with code allowable (e.g. for $D = 3$ m, $243.7/(\pi \times 3) = 25.9$ kN/m, a routine fillet-weld duty).

5. Major support beam (3 m embodiment). A diametral truss of tributary width $b_t = 0.5$ m carries UDL $w = qb_t = 17,237$ N/m over span $L = 3$ m: $M = wL^2/8 = 19.4$ kN·m; required section modulus $Z = M/\sigma_{allow} = 1.39 \times 10^{-4}$ m³ (139 cm³), met by a standard truss or channel.

6. Dynamic amplification. A suddenly applied surge gives a dynamic amplification factor up to DAF = 2. Design practice: either (a) size the static structure for $2\times$ the relieved sustained load, or (b) rely on the relief flaps to limit the impulse. AEGIS uses both: ribs/clamps sized for the 5 psi peak with DAF, flaps capping the sustained load.

7. Relief-flap sizing (parametric). Flaps of total area A_f crack at set pressure ΔP_{set} (set by flap mass per unit area plus any spring: $\Delta P_{set} = m_f g/A_f + kx_0/A_f$). Once open, they pass the surge vapour at

$$Q_{relief} = C_d A_f \sqrt{2 \Delta P / \rho_V}.$$

Sizing A_f to relieve the credible surge volumetric rate at $\Delta P \leq \Delta P_{set} (1 + \text{margin})$ caps the sustained deck load near ΔP_{set} (Fig. 2c). A flap area of 5–10% of tray area typically suffices; flaps reseal under their own weight as the surge decays, restoring normal operation. *Normal-operation seal:* at design vapour the upward force on a seated flap ($\Delta P_{tray} \sim 6$ mbar) is far below ΔP_{set} (~ 140 mbar at 2 psi), so flaps stay shut and contribute nothing to normal hydraulics.

8. Free-area vs stiffness trade-off (parametric). Adding ribs and clamps consumes active area and adds weight. The optimisation maximises bubbling area fraction ϕ_a subject to $\sigma(s, t) \leq \sigma_{allow}$ and panel mass ≤ 25 kg (manway handling). For the chosen 2.5 mm/170 mm design, rib footprint costs $<4\%$ of bubbling area, so normal capacity and efficiency are essentially unchanged.

9. Operating window. Because the relief flaps are seated in normal operation, the jet-flood, weep, downcomer-choke and downcomer-backup boundaries are identical to the underlying valve/sieve tray. AEGIS adds a new *upper survival boundary* far above the normal flood line: the tray survives excursions that would destroy a conventional tray, rather than extending the steady operating range.

7 Design Parameter Envelope (claim-band table)

Parameter	Claimed band	Preferred
Uplift structural rating	3–5 psi (21–34.5 kPa)	5 psi
Deck thickness	2.0–3.0 mm SS	2.5 mm
Rib pitch s	$\leq t\sqrt{\sigma_{allow}/0.75q}$	150–180 mm
Allowable stress σ_{allow}	$\leq \text{yield}/1.5$	140 MPa
Hook-clamp pitch	150–250 mm	200 mm
Clamp load (at 5 psi, $b=0.4$ m)	—	~ 1.4 kN each
Relief flap set pressure	1.5–3 psi	2 psi
Relief flap total area	5–10 % of tray area	7 %
Dynamic amplification factor	1.5–2.0	2.0
Bubbling-area loss vs standard	$<5\%$	$<4\%$

Table 1: Claim-band table; ranges are dependent-claim fallback positions.

Metric	Conventional clamped tray	Heavy bolt-down tray	AEGIS
Uplift rating	0.15–0.3 psi	1–2 psi	5 psi
Deck stress at 5 psi (2.5 mm, 170 mm pitch)	buckles	—	~155 MPa
Manway removable	yes	no	yes
Panel mass	~18 kg	~35 kg	~22 kg
Normal ΔP / efficiency / turndown	base	base	= base
Surge survival (relief-capped)	no	marginal	yes

Table 2: Embodiment values from the calculations above; pending FEA confirmation.

8 Predicted Performance (embodiment matrix)

9 Claims (draft)

Claim 1. A cross-flow contacting tray comprising a welded support ring, deck panels each locked to said ring by interlocking hook clamps engaging the ring and resisting vapour uplift in tension, and stiffening ribs on the panel undersides at a pitch s no greater than $t\sqrt{\sigma_{allow}/0.75q}$ for deck thickness t , allowable stress σ_{allow} and design uplift pressure q of at least 21 kPa, whereby the tray resists a vapour uplift of at least 3 psi without panel dislodgement or buckling.

Claim 2. The tray of claim 1, further comprising integral relief flaps that are seated and sealed during normal operation and crack open at a set differential pressure of 1.5–3 psi to vent vapour and cap the sustained uplift load on the deck.

Claim 3. The tray of claim 2, wherein the relief flaps have a combined area of 5–10% of the tray area and reseal under their own weight upon decay of the surge.

Claim 4. The tray of claim 1, rated for a vapour uplift of 5 psi (34.5 kPa).

Claim 5. The tray of claim 1, wherein the hook clamps are at a pitch of 150–250 mm and each panel remains removable through a column manway.

Claim 6. The tray of claim 2, wherein the relief flaps are sealed at design vapour load because the design tray pressure drop is below 10% of the flap set pressure, so that normal-operation hydraulics, efficiency and turndown are unchanged relative to a tray without said flaps.

Claim 7. The tray of claim 1, wherein the deck is a valve, sieve, fixed-valve or adaptive-flap deck, the uplift-resistance features being superimposed on said deck without reducing bubbling area by more than 5%.

Claim 8. A method of designing a surge-tolerant tray, comprising specifying a design uplift pressure q ; selecting rib pitch from $s = t\sqrt{\sigma_{allow}/0.75q}$; sizing hook clamps for a tensile line load $qb/2$; and selecting relief-flap area and set pressure to cap the sustained deck load below the structural rating.

Claim 9. A column containing trays according to any preceding claim in a service subject to pressure-surge, relief, slug-flow or steam-out uplift transients.

10 Industrial Applicability

Columns subject to surge: those downstream of relief headers, fired reboilers prone to bumping, slug-flow feeds, rapid-depressurisation (blowdown) service, and units with a history of tray collapse;

also a general reliability upgrade where a single tray-collapse incident outweighs the modest added cost.

Prior-art neighbours to clear: tray hold-down clamp and bar art, bolted/welded tray attachment, anti-uplift valve units, and any “relief tray”/blow-off panel art. The asserted inventive step is the integrated system: positive interlock + computed rib stiffening + integral resealing relief flaps that bound the structural duty while leaving normal hydraulics untouched, with the parametric design method of claim 8.